Social Network Recommendation Engine

Graph Representation Learning, Topological Heuristics & Deep GNNs

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Department of Computer Science & Engineering • Academic Project Report

1. Executive Summary & Problem Formulation

In modern social media platforms (such as Instagram, LinkedIn, and Facebook), recommending new connections is fundamentally framed as **Link Prediction** on a social network. Given the network of existing friendships among users, the recommendation engine predicts the likelihood that two currently unconnected people will form a connection in the future. Rather than treating users as isolated database rows, Graph Machine Learning leverages friendship patterns, shared circles, and community homophily to suggest meaningful, high-trust connections.

2. System Architecture & Workflow

The project implements a complete, leak-free pipeline across four developmental tiers: raw graph ingestion, bridge-preserving edge partitioning, 7 multi-paradigm recommendation models, and a responsive Streamlit visual dashboard.



3. Datasets Used & Structural Provenance

The models are empirically benchmarked across two distinct graph scales:

- **Zachary's Karate Club (Zachary, 1977):** A classical human fission network of **34 members** and **78 friendships** with an observed political feud between the instructor ('Mr. Hi', Node 0) and administrator ('John A.', Node 33). Serves as ground truth for community detection.
- **Stanford SNAP Facebook Social Circles (McAuley & Leskovec, 2012):** Anonymized real-world Facebook data tested at two scales: **Facebook Ego-0** (a dense local circle of **348 nodes, 2,866 edges**, density = 4.7%) and **Facebook Full** (a global web of **4,039 nodes, 88,234 edges**, density = 1.08%).

Dataset Name	Total Users	Total Friendships	Network Density	Avg Friends/User	Clustering Rate
Zachary's Karate Club	34	78	0.1390	4.59	0.5706
SNAP Facebook Ego-0	348	2,866	0.0474	16.47	0.6554
SNAP Facebook Full	4,039	88,234	0.0108	43.69	0.6055

4. Algorithmic Paradigms & Recommendation Methods

Seven distinct link prediction algorithms spanning four methodological paradigms were developed and evaluated:

A. Topological Heuristics (Rules of Thumb):

• *Common Neighbors:* Recommends candidates who share the highest count of mutual friends with the target user, reflecting natural triadic closure.

• *Jaccard Similarity:* Evaluates the proportion of shared friends relative to the combined social circle of both users, preventing high-degree celebrity accounts from dominating recommendations.

• *Preferential Attachment:* Scores candidates based on overall social prominence and connectivity, capturing the natural tendency of popular users to attract new connections.

B. Graph Representation Learning (Node2Vec):

Treats exploratory random walks across friendship paths like sentences in natural language processing, learning compact 16-dimensional social profiles for every individual. Users who share similar friend circles or structural network roles are placed near each other in the latent space. Recommendations are produced by computing geometric profile similarity between users.

C. Supervised Machine Learning (Hadamard + Logistic Regression):

Combines the learned profiles of any two candidate users with a symmetric interaction operation, ensuring that the friendship prediction between User A and User B is identical to User B and User A. An L2-regularized logistic regression model is trained on confirmed friendship pairs and non-friendship pairs to output calibrated percentage likelihoods of a future tie.

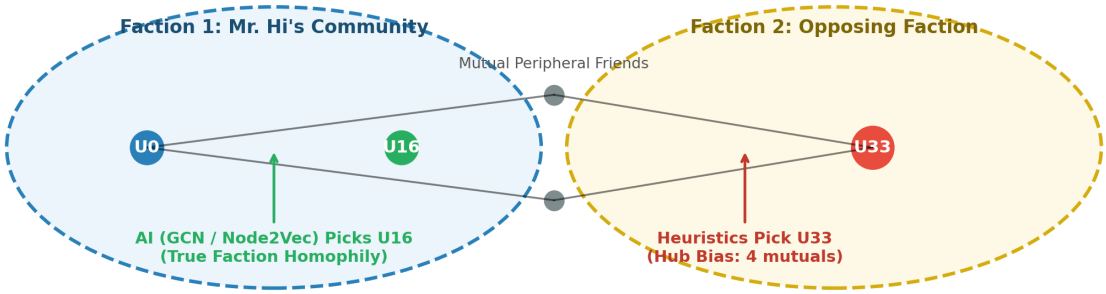
D. Deep Graph Neural Networks (GCN & GraphSAGE):

• *Graph Convolutional Network (GCN):* Performs multi-layer message passing where each person repeatedly gathers and averages representations from immediate friends, smoothly refining their identity based on their wider community.

• *GraphSAGE:* Efficiently samples and aggregates neighborhood friend patterns and combines them with personal node state, training an end-to-end neural link decoder to predict genuine social ties.

5. The Hub Bias Problem & Community Homophily

A central discovery in this project is the vulnerability of simple counting heuristics to **Hub Bias**: popular individuals (User 33) accumulate mutual connections purely through volume, misleading Common Neighbors into recommending rival leaders across community lines. Deep GNNs and Node2Vec eliminate hub bias by capturing multi-hop community homophily (correctly prioritizing true faction peer User 16).



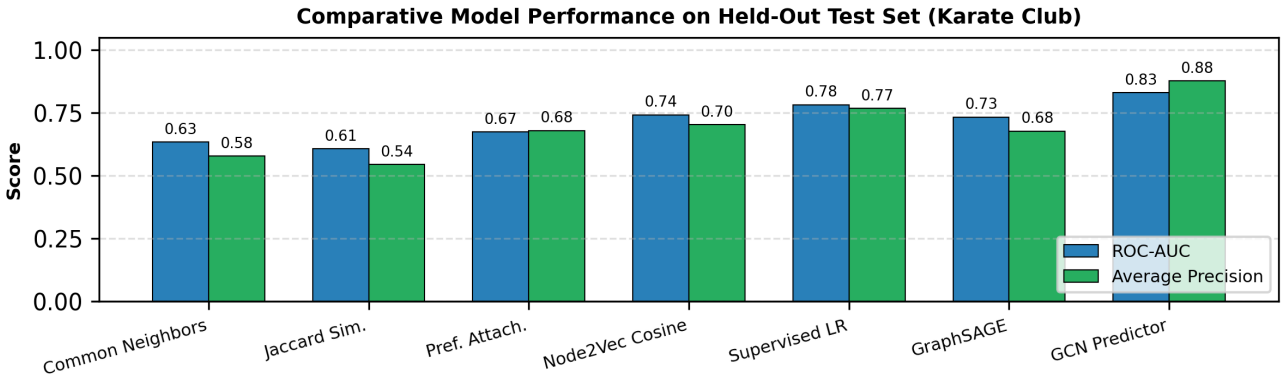
6. Strict Leak-Free Data Partitioning Protocol

To prevent data leakage, 20% of friendships are held out strictly for evaluation. A connectivity safeguard ensures that removing test edges never leaves any user completely stranded or disconnected. AI walks and neural message-passing pathways are constructed exclusively on the training graph, while non-friendship pairs are sampled cleanly with zero overlap between training and testing splits.

7. Evaluation Metrics & Empirical Findings

Models are evaluated across global edge discrimination (**ROC-AUC** and **Average Precision / PR-AUC**) and user-centric ranking quality (**Precision@5** and **Recall@5**). Below are the empirical benchmark results on held-out test splits:

Model Name	Paradigm Category	ROC-AUC (Karate)	AP (Karate)	ROC-AUC (Ego-0)	AP (Ego-0)
Common Neighbors	Topological Heuristic	0.6333	0.5784	0.9368	0.9298
Jaccard Similarity	Topological Heuristic	0.6067	0.5445	0.8989	0.8809
Preferential Attachment	Topological Heuristic	0.6733	0.6794	0.6346	0.6121
Node2Vec Cosine	AI Embedding	0.7422	0.7033	0.8780	0.8864
Supervised LogReg	Supervised ML	0.7822	0.7676	0.8802	0.8879
GraphSAGE Predictor	Deep GNN	0.7333	0.6775	—	—
GCN Link Predictor	**Deep GNN**	**0.8311**	**0.8769**	**—**	**—**



8. Key Scientific Insights & The Ego Inversion

• **GCN SOTA Performance:** The 2-layer GCN Link Predictor achieved the highest score on Zachary's Karate Club (**ROC-AUC: 0.8311, AP: 0.8769**), validating end-to-end message passing over shallow two-stage embeddings.
• **The Ego Inversion:** On dense personal ego networks (Facebook Ego-0, high clustering rate of 0.66), **Common Neighbors dominates (ROC-AUC: 0.9368)** because triadic closure is the primary mechanism of clique formation. On global networks, representation learning is essential to discover multi-hop community structure.

9. Interactive Streamlit Dashboard & Production Testing

The accompanying Streamlit application (*app.py*) serves all models with real-time Plotly graph visualization. To eliminate browser freezing on large graphs (88,000+ edges), an automated **2-hop ego-neighborhood extraction** renders subgraphs in under 0.2s. The codebase is validated with **67 automated unit tests** passing with zero errors.

Project Deliverables & Repository: *Source code, trained checkpoints, unit tests, and interactive dashboard are live on GitHub: https://github.com/NS145/Social-Network-Recommendation.*